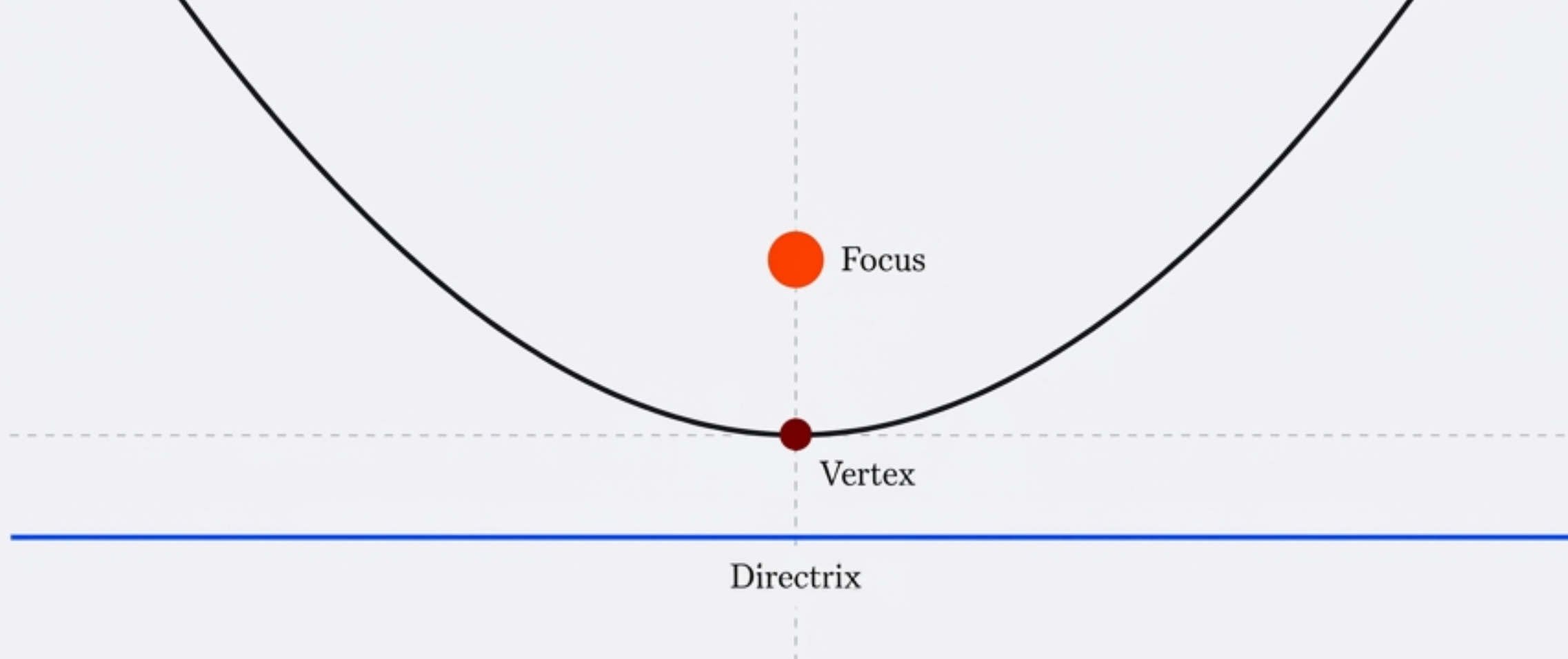


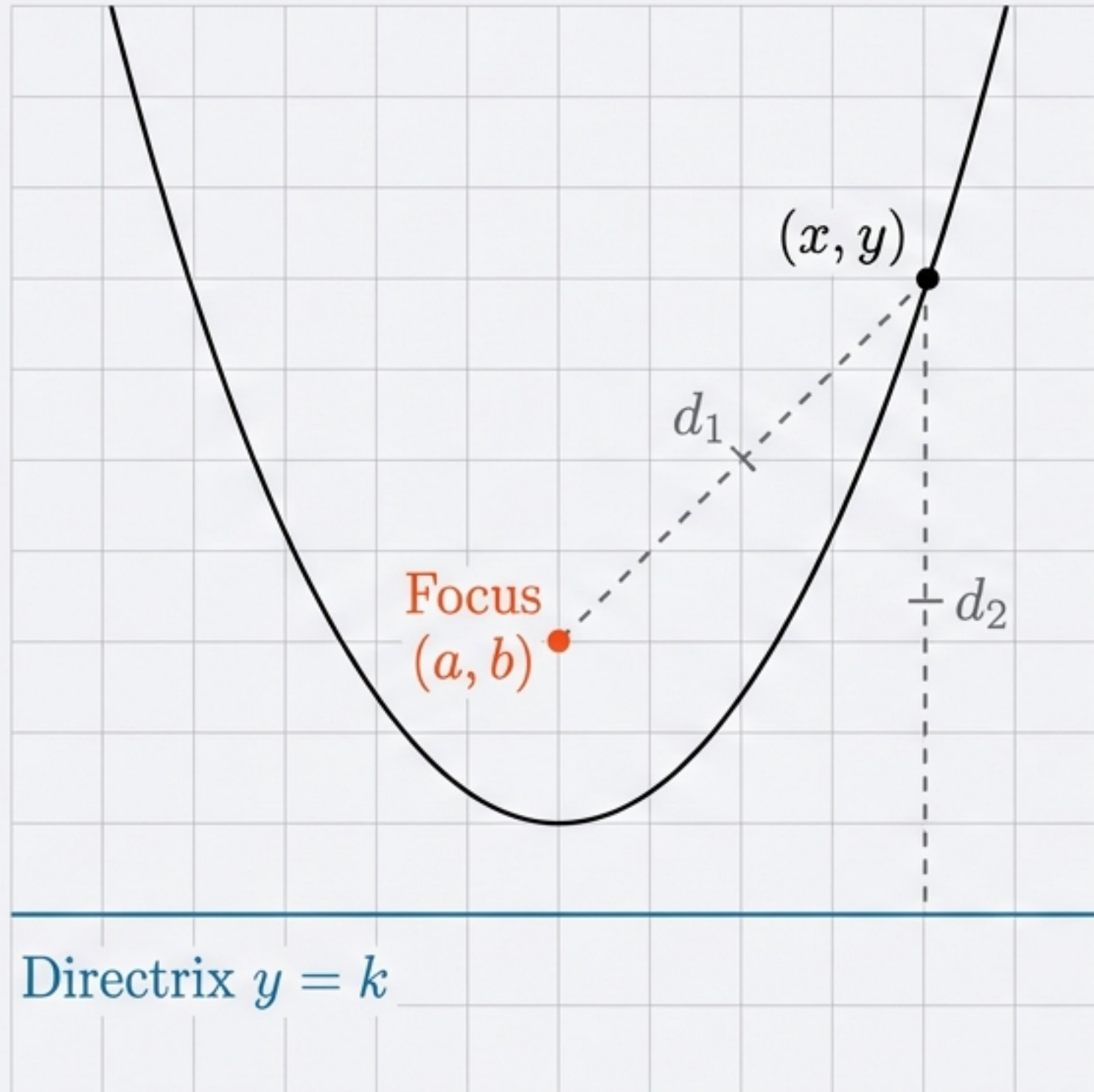
The Hidden Geometry of Parabolas

Decoding the relationship between the Vertex, the Focus, and the Directrix.



A visual guide to deriving the Golden Rule of parabolic distance: $1 / 4A$

A Parabola is a Perfect Balance



The Definition:

A parabola is the 'Locus of Points' where every single point on the curve is perfectly equidistant from a fixed point (the Focus) and a fixed line (the Directrix).

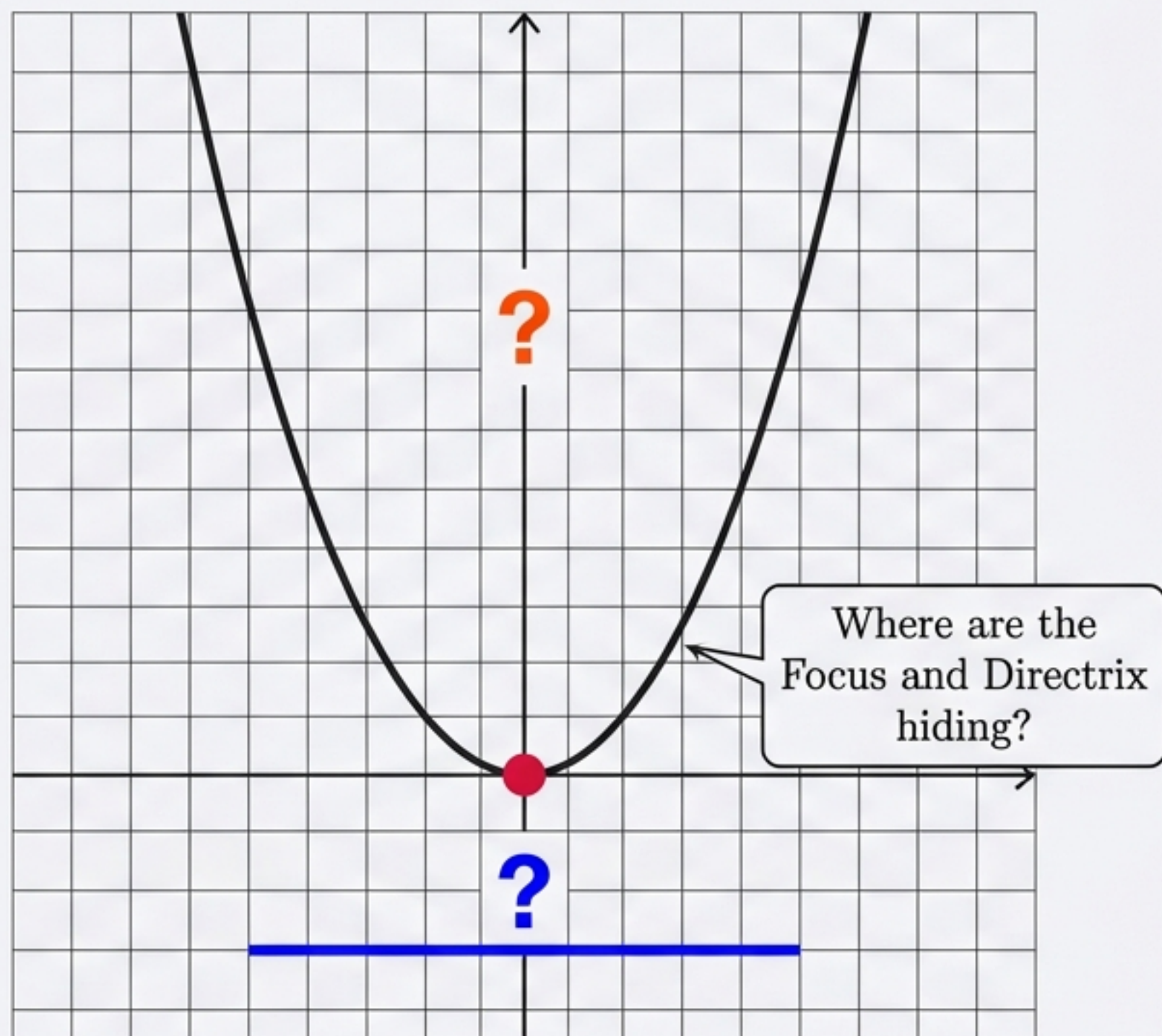
The Equation:

$$d_1 = d_2$$

The Objective:

Previously, we used the distance formula to build the curve from these points. Now, we will reverse-engineer the process: starting with the curve to find the hidden geometry.

The Prototype: Analyzing $y = x^2$

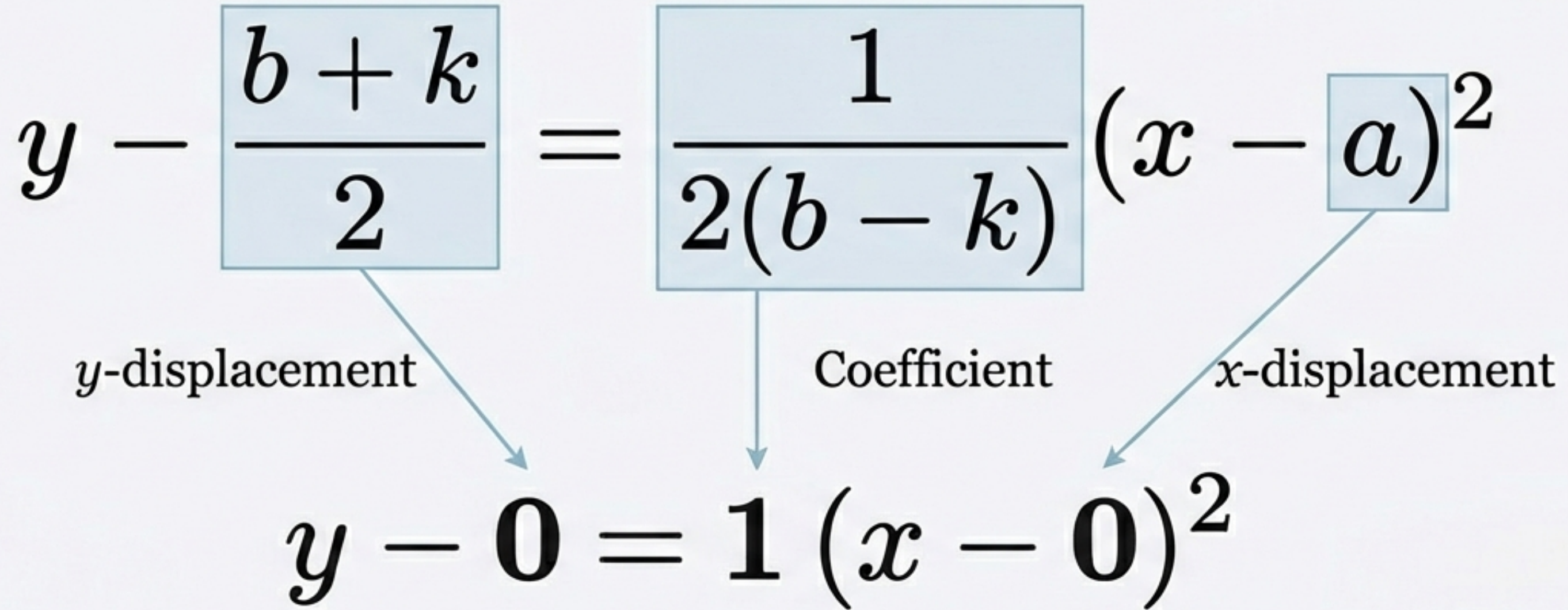


We know the curve. Now we must find the machinery that created it. To do this, we rewrite the equation to expose its full structure:

$$y - 0 = 1(x - 0)^2$$

- Vertex (x_1, y_1) : $(0, 0)$
- Coefficient A : 1
- Analysis: Since A is positive, the curve opens upward. The Focus is on the axis of symmetry (y -axis), and the Directrix is a horizontal line below.

Mapping the Variables


$$y - \frac{b+k}{2} = \frac{1}{2(b-k)} (x - a)^2$$

y -displacement Coefficient x -displacement

$$y - \mathbf{0} = \mathbf{1} (x - \mathbf{0})^2$$

By visually matching these patterns, we can extract a system of equations to solve for b (Focus height) and k (Directrix height).

Solving for the Hidden Coordinates

Set Up

$$\frac{b+k}{2} = 0 \rightarrow b+k=0$$
$$\frac{1}{2(b-k)} = 1 \rightarrow b-k = \frac{1}{2}$$

Solve

Add the two equations together:

$$(b+k) + (b-k) = 0 + \frac{1}{2}$$
$$2b = \frac{1}{2}$$

$$b = \frac{1}{4}$$

Substitute back to find k :

Since $b+k=0$, then...

$$k = -\frac{1}{4}$$

Result

We have located the geometry.

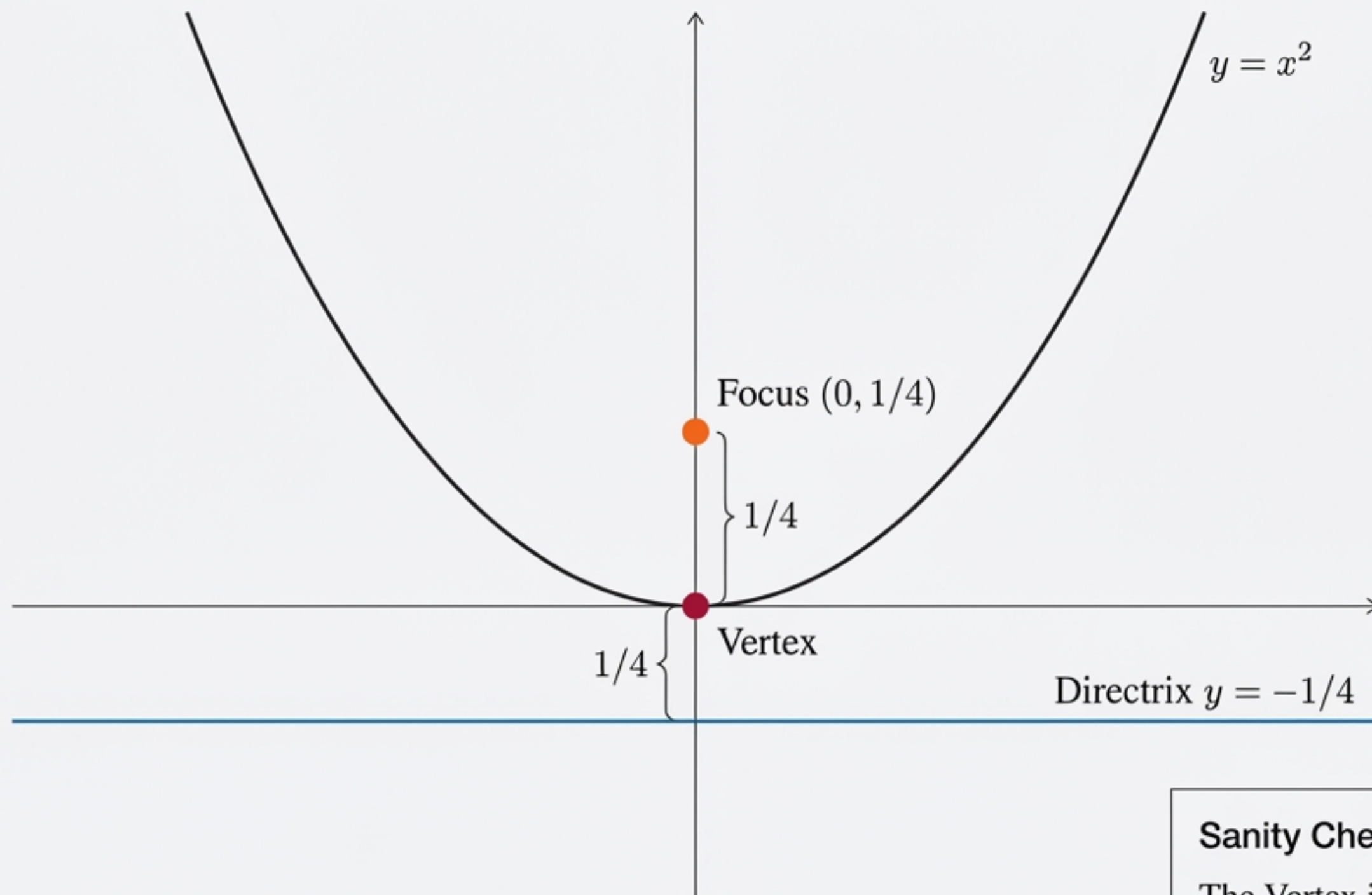
Focus Coordinate

$$\left(0, \frac{1}{4}\right)$$

Directrix Equation

$$y = -\frac{1}{4}$$

The Geometry of $y = x^2$ Revealed



Sanity Check:

The Vertex is exactly in the middle.
The Focus is 'inside' the cup of the curve.
The geometry holds.

Generalizing the Pattern

$$y - y_1 = A(x - x_1)^2$$

The Vertex
(The anchor point).

The Scaling Factor
(Determines steepness).

We don't want to solve a system of linear equations every time. We need a universal rule that works for *any* scaling factor A , not just when $A = 1$. We need to solve for the relationship between A and the Focus.

Setting up the General System

The Pattern Match

$$\frac{1}{2(\textcolor{red}{b} - \textcolor{blue}{k})} = A$$

Focus ↓

$$\frac{\textcolor{red}{b} + \textcolor{blue}{k}}{2} = y_1$$

↑ Focus Directrix ↑

The Rearrangement

We need to solve for b (Focus) and k (Directrix) using only A and y_1 .

Distance Equation

$$\textcolor{red}{b} - \textcolor{blue}{k} = \frac{1}{2A}$$

Midpoint Equation

$$\textcolor{red}{b} + \textcolor{blue}{k} = 2y_1$$

Deriving the Focus Coordinate

Step 1: Start with our system:

$$\begin{array}{l} \rightarrow b + k = 2y_1 \\ \rightarrow b - k = \frac{1}{2A} \end{array}$$

Step 2: Add the equations together to eliminate k :

$$2b = 2y_1 + \frac{1}{2A}$$

Step 3: Divide by 2 to isolate b (The Focus y-coordinate):

$$b = y_1 + \frac{1}{4A}$$

Translation:

The Focus is simply the Vertex height (y_1) **PLUS** a specific distance: $\frac{1}{4A}$.

Deriving the Directrix Line

Step 1: Recall the equidistant rule:
Distance from Focus to Vertex =
Distance from Vertex to Directrix.

Step 2: Or substitute b back into our
midpoint equation:

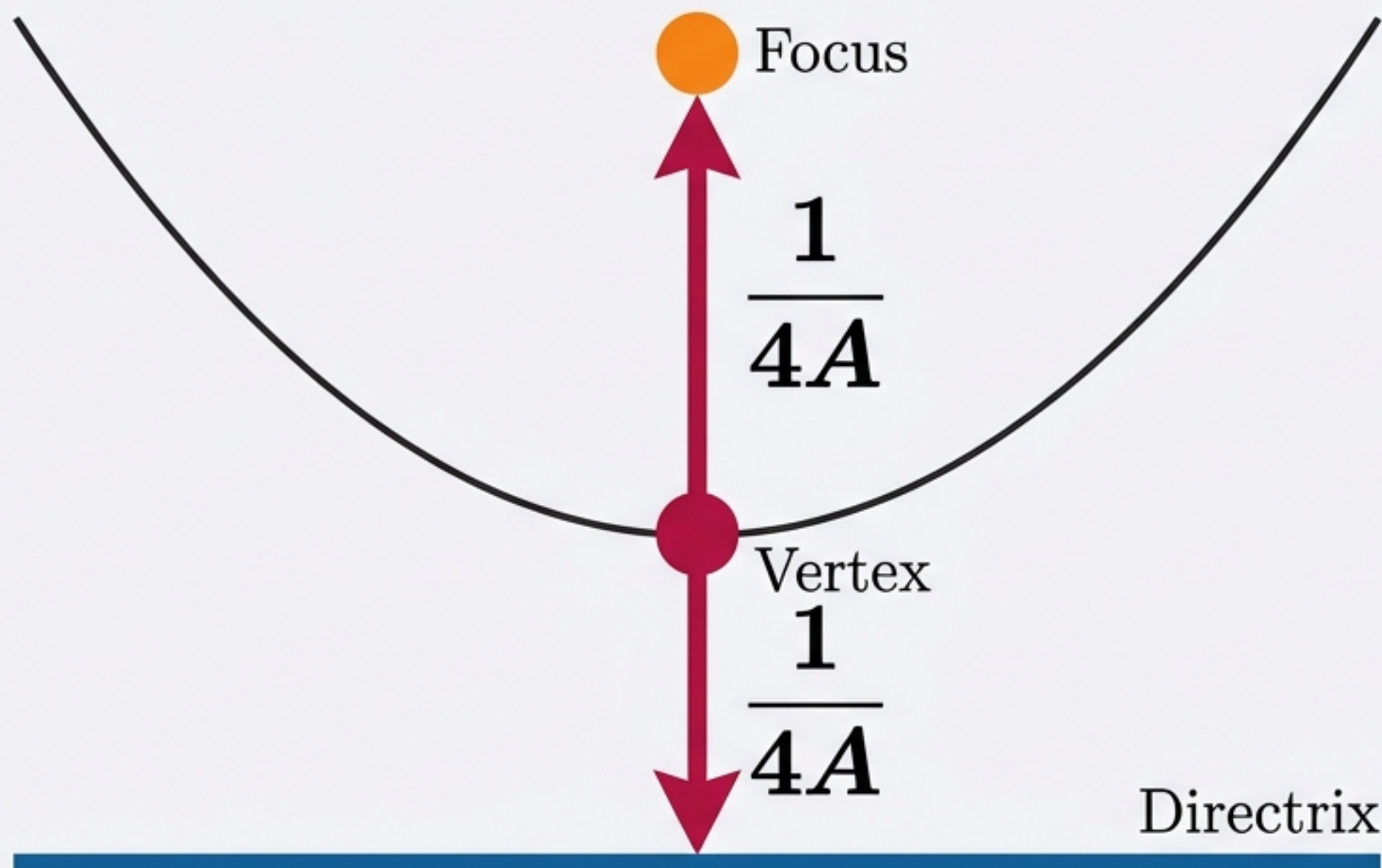
$$2y_1 = \left(y_1 + \frac{1}{4A}\right) + k$$

Step 3: Solve for k
(The Directrix y -coordinate):

$$k = y_1 - \frac{1}{4A}$$

Translation: The Directrix is simply the Vertex height (y_1) **MINUS** that same distance: $\frac{1}{4A}$.

The Golden Rule of Parabolas



The Key Insight

The distance from the Vertex to the Focus is always determined by the scaling factor alone.

$$\text{Distance} = \frac{1}{4A}$$

This simple fraction unlocks the geometry of any parabola written in vertex form.

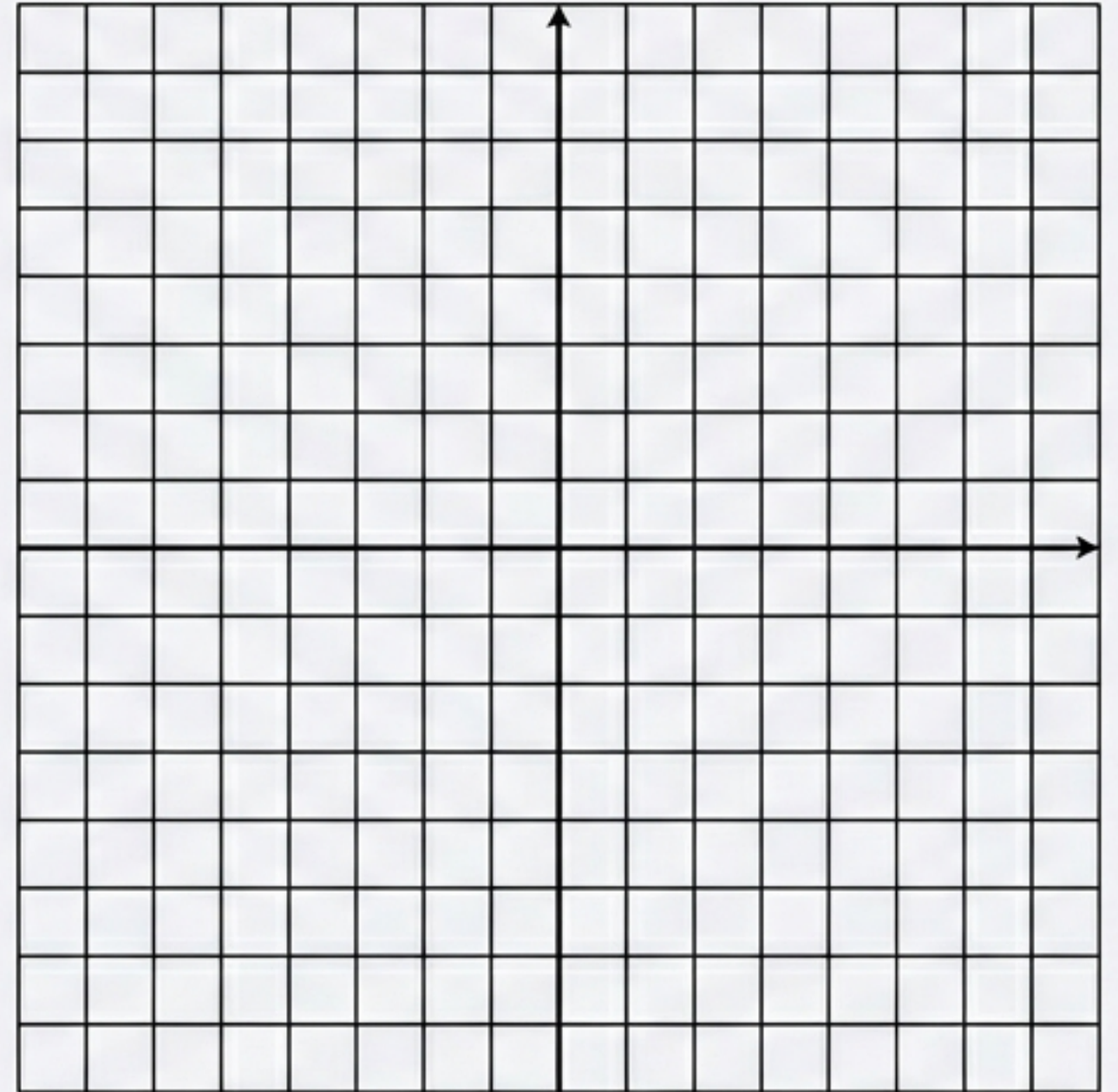
The Application Challenge

$$y - 1 = 2(x - 3)^2$$

Challenge: Find the Focus and Directrix.

This looks complicated. Do we need to set up a system of linear equations again?

No. We use the Golden Rule.



Inspection, Not Algebra

The Analysis

$$y - 1 = 2(x - 3)^2$$

Vertex (3, 1)

$A = 2$

The Golden Rule Calculation

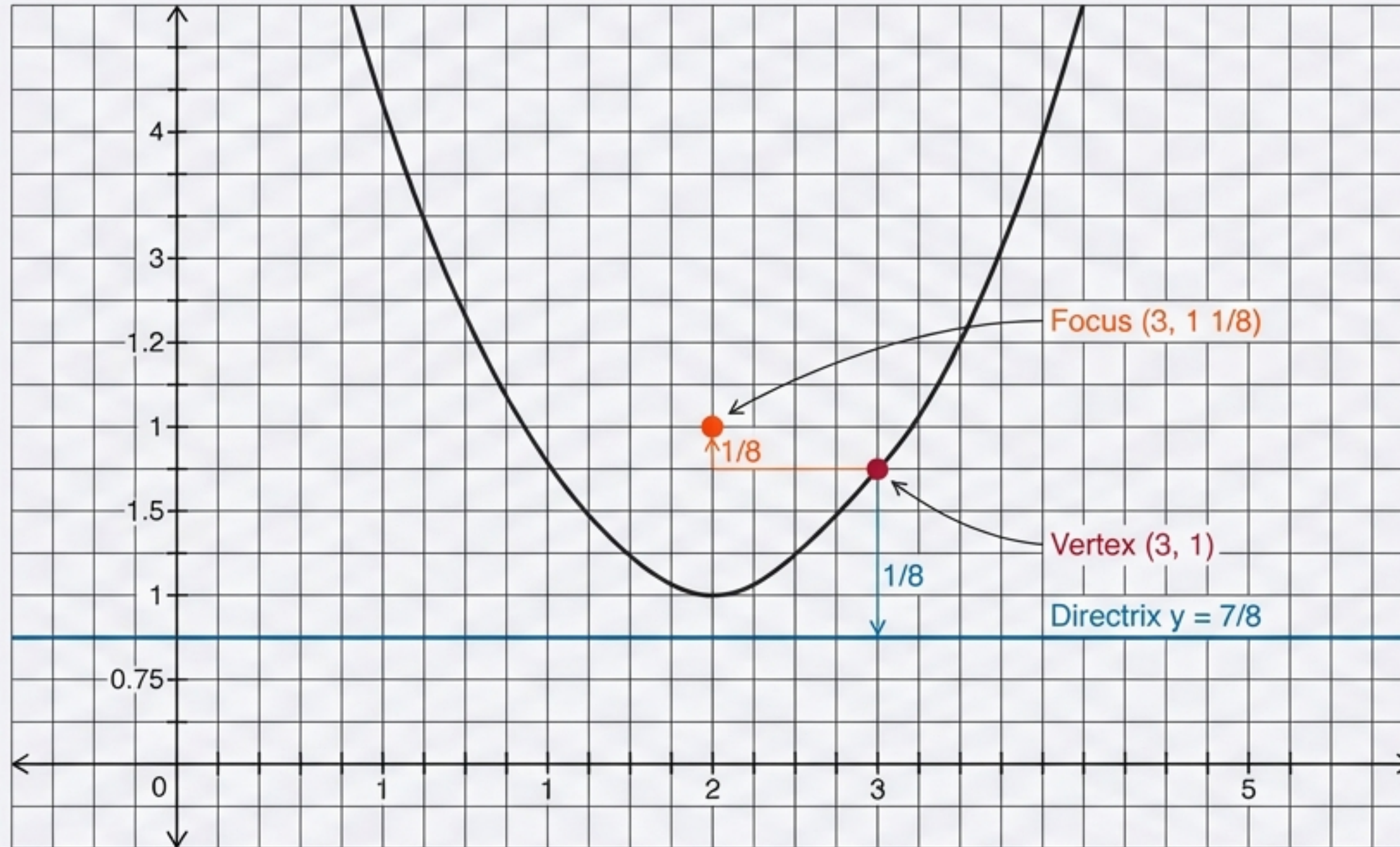
Calculate the distance:

$$\text{Distance} = \frac{1}{4A} = \frac{1}{4(2)} = \frac{1}{8}$$

The Plan:

1. Start at Vertex (3, 1).
2. Go UP $1/8$ to find the Focus.
3. Go DOWN $1/8$ to find the Directrix.

The Solved Geometry



By inspection alone, we have defined the curve's entire structure.

Parabola Anatomy Cheat Sheet

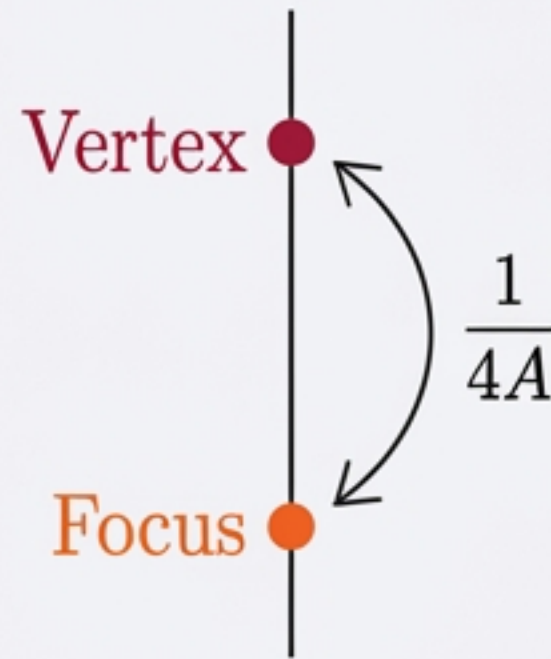
Vertex Form

$$y - y_1 = A(x - x_1)^2$$

A determines the distance.

The Key Metric

$$\text{Distance} = \frac{1}{4A}$$



Finding Points

Start at Vertex (x_1, y_1) .

Focus

$$\left(x_1, y_1 + \frac{1}{4A}\right)$$

Directrix

$$y = y_1 - \frac{1}{4A}$$

With the equation in vertex form, finding the hidden geometry is a matter of inspection.